MILITARY SPECIFICATION SHEET

SCANNER, MECHANICAL, MX-10717(V)3/U

This specification is approved for use by USACECOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification sheet covers the Mechanical Scanner, MX-10717(V)3/U and augments MIL-S-49301. The MX-10717(V)3/U is used on the AN/VSG-2, AN/VSG-2A, Tank Mounted Thermal Sight.

1.2 Purpose. The purpose of this specification sheet is to modify or expand the contents of MIL-S-49301 to describe the MX-10717(V)3/U Mechanical Scanner. All the MIL-S-49301 contents apply except as specifically modified herein. Paragraph numbers directly correlate between MIL-S-49301 and this specification sheet.

2. APPLICABLE DOCUMENTS

2.1 Order of precedence. In the event of a conflict between the text of this specification sheet and the references cited herein, the text of this specification sheet shall take precedence.

SPECIFICATIONS

MILITARY

MIL-S-49301	- Scanner, Mechanical, MX-10717(V)/U, General Specification for

DRAWINGS

USA COMMUNICATIONS-ELECTRONICS COMMAND

54490 5006243-3	- Scanner, Mechanical, MX-10717(V)3/U
80063 SM-D-773946	- Scan-Interlace, PL-1408/UA

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-ED-TO, Fort Monmouth, NJ 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5855

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3. REQUIREMENTS

3.1 Description. The MX-10717(V)3/U Scanner, herein known as the MX-10717(V)3/U, shall consist of the MIL-S-49301, MX-10717(V)/U Scanner with a phase-shift lens and counterweight assembled and adjusted for 42 Hz operation.

3.2 Construction. The MX-10717(V)3/U shall be constructed in accordance with drawing 54490 5006243-3.

3.2.1 Weight. Weight of the MX-10717(V)3/U shall be 1.55 pounds maximum.

3.5 Components.

a. Scanner, Mechanical MX-10717(V)/U
b. Phase-shift lens assembly
c. Counterweight

3.6 Performance characteristics. The characteristics specified herein shall apply as defined in 3.1 and 3.6.1 except 4.6.8, 4.7.4, and 4.7.5 apply with the phase shift lens, counterweight and associated hardware removed. Operation shall be at 42 Hz using the Scan-Interlace, PL-1408/UA (SM-D-773946).

3.6.1 Scan angle and frequency.

3.6.1.1 Variable scan frequency. Not applicable.

3.6.1.2 Specific scan frequency (42 Hz). Scan frequency of the MX-10717(V)3/U shall be adjusted to 42.0, ±1.0 Hz. Scan frequency drift shall not be greater than ±4.2 Hz from the frequency at ambient setting (see MIL-S-XXXXX, 4.4) when exposed to high and low temperature environmental conditions (see MIL-S-49301).

3.6.1.3 Synchronous scan frequency. Not applicable.

3.6.1.4 Scan angle. The scan angle shall be 12.8, ± 0.5 degrees.

3.6.2 Interlace. The MX-10717(V)3/U shall operate in the 2:1 interlace mode. The interlace switching time shown in MIL-S-49301, figure 1, shall not be greater than 3.6 milliseconds (ms).

3.6.3 Scan direction. The MX-10717(V)3/U shall show the waveform shapes and phase relationships in figure 1 while operating in the 2:1 rear interlace mode.

3.6.4 Power consumption. Power required to operate the MX-10717(V)3/U shall not exceed 7.5 watts.

3.6.5 Scan efficiency. The MX-10717(V)3/U scan efficiency shall be a minimum of 65 percent.

3.6.6 Scan jitter in azimuth. Jitter time shall be 18.0 microseconds maximum.

3.7 Environmental conditions. See MIL-S-49301 except as follows:

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3.7.5 Vibration. The MX-10717(V)3/U shall not be damaged by vibration over a frequency spectrum at the specified g levels and amplitudes shown in figure 2 for the X axis and as shown in figure 3 of MIL-S-49301 for the Y and Z axes. During specified vibration, the average interlace waveform amplitude shown in MIL-S-49301, figure 1 shall not deviate from the amplitude without vibration by more than ±50 percent.

4. QUALITY ASSURANCE PROVISIONS

4.6 Test methods. Tests shall be performed utilizing the SM-D-773946 Scan-Interlace with the MX-10717(V)3/U with a scan angle of 12.8, ±0.5 degrees at the 42.0, ±1.0 Hz scan rate.

4.6.1 Scan frequency.

4.6.1.1 Variable scan frequency. Not applicable.

4.6.1.2 Specific scan frequency (42 Hz). The MX-10717(V)3/U shall be connected to the Scan-Interlace SM-D-773946 and mounted to a suitable fixture. Power shall be applied and the output of the azimuth (scan) transducer (Scan-Interlace SM-D-773946) shall be monitored with a calibrated frequency counter. The output of the Scan-Interlace (SM-D-773946) shall be adjusted to obtain the specific scan frequency. Failure to meet the requirements of 3.6.1.2 shall constitute failure of this test.

4.6.1.3 Synchronous scan frequency. Not applicable.

4.6.1.4 Scan angle. See MIL-S-49301.

4.6.2 Interlace. See MIL-S-49301.

4.6.3 Scan direction. With the MX-10717(V)3/U operating, the outputs of the azimuth and interlace position transducers (Scan-Interlace SM-D-773946) shall be observed with an oscilloscope. The output waveforms shall show the shape and phase relationships shown in figure 1. Failure to meet the requirements of 3.6.3 shall constitute failure of this test.

4.6.4 Power consumption. See MIL-S-49301.

4.6.5 Scan efficiency. See MIL-S-49301.

4.6.6 Scan jitter in azimuth. See MIL-S-49301.

4.7 Environmental tests. See MIL-S-49301 and figure 2.

6. NOTES

6.1 Intended use. The MX-10717(V)3/U is intended for use with the AN/VSG-2 and AN/VSG-2A Tank Mounted Thermal Sight.

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6.2 Interchangeability. The MX-10717(V)3/U scanner built and tested in accordance with this specification and applicable drawings may be used in place of the MX-9872(V)3/UA scanner. The MX-9872(V)3/UA may not be used in place of the MX-10717(V)3/U scanner.

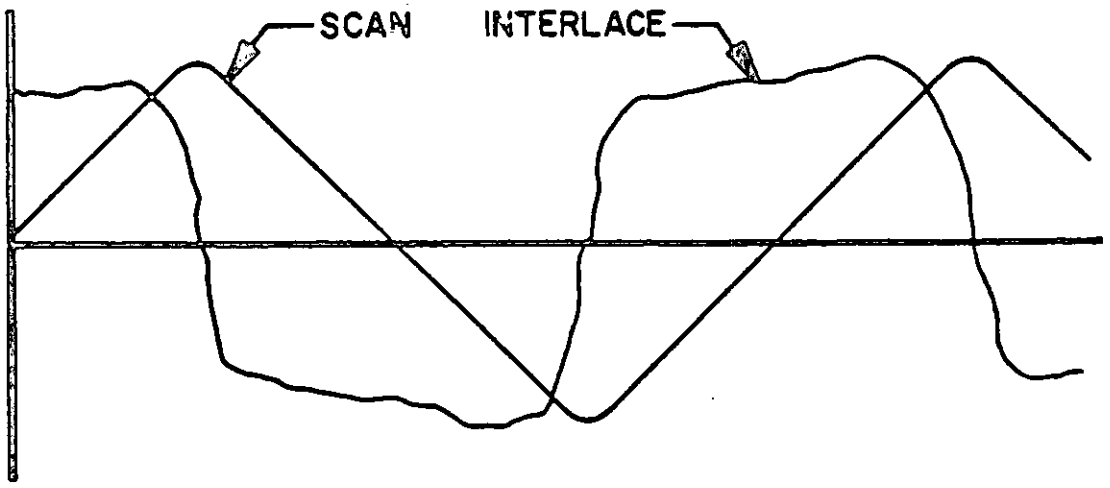
6.3 Subject Term (Keyword) Listing).
FLIR
Sensor
Scanner

CUSTODIAN:
Army - CR

PREPARING ACTIVITY
Army - CR
Project 5855-A290-3

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2:1 REAR POSITION
INTERLACE LEADS SCAN APPROX. 90°



NOTES:

PL-1408/UA- CONNECT P1-33 TO P1-34 AND P1-4 TO P1-17

OSCILLOSCOPE- CONNECT CHANNEL 1 (SCAN POSITION) HIGH INPUT TO J3 (OR J2-7) AND LOW SCOPE TO GND

- CONNECT CHANNEL 2 (SCAN INTERLACE) HIGH INPUT TO J6 (OR J1-9) AND LOW SCOPE TO GND

FIGURE 1. INTERLACE TRANSDUCER AND AZIMUTH POSITION TRANSDUCER OUTPUTS.

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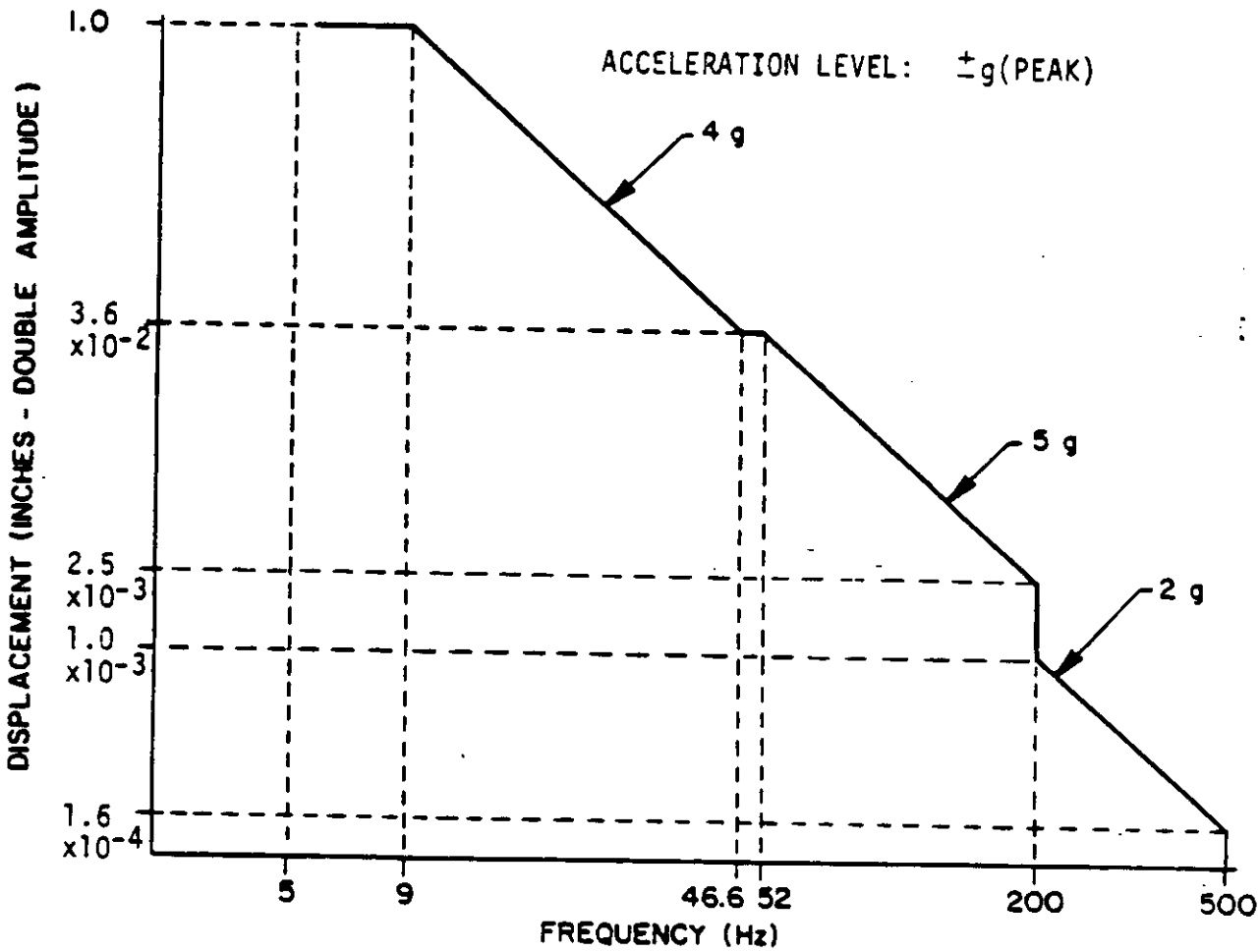


FIGURE 2. VIBRATION TEST PROFILE (X AXIS).

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-49301/3(CR)	Scanner, Mechanical, MX-10717(V)3/U

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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